

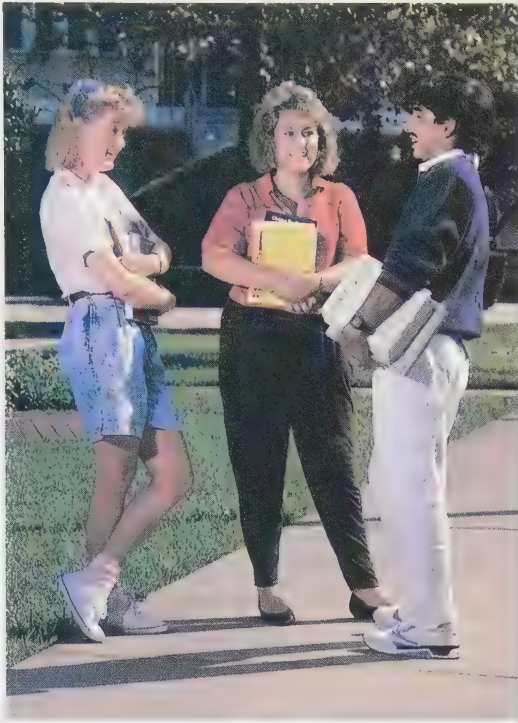


FIGURE 10.36 One of many poorly understood differences between the sexes: Beyond the age of puberty, most males carry packages at their side, whereas females carry them in an elevated position.

could say that women are more conformist or more flexible and that men are more independent or more stubborn.) Males and females also differ in helping behavior. It is not a simple case of one sex being more helpful than the other; men and women tend to help in different ways. For example, men are generally more likely to help a stranger change a flat tire; women are more likely to help people who need long-term nurturing support (Eagly & Crowley, 1986).

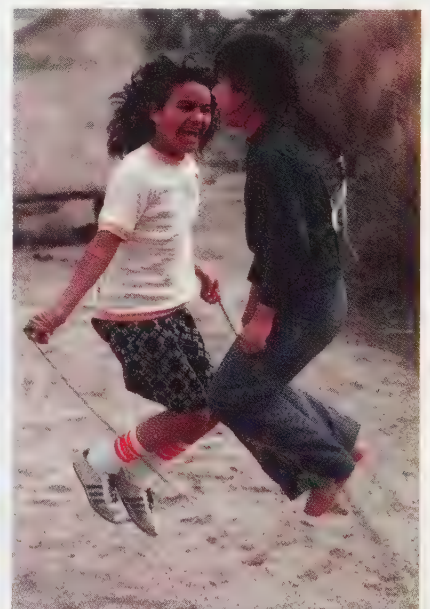
Sex Differences in Social Situations It is now clear that certain important differences between males and females emerge only in a social context (Maccoby, 1990). Psychologists ordinarily test people in isolation. When tested one at a time, boys and girls tend to behave about the same in most regards. However, in a group setting, boys usually get together with other boys and girls get together with other girls; suddenly, the two groups act very differently (Figure 10.37).

Girls sometimes play competitive games, but they are more likely than boys to spend long times at quiet, cooperative play. They take turns; they present their desires as “suggestions” instead of demands; they exchange compliments; and they generally try to avoid hurting each other’s feelings.

Meanwhile, boys are almost always competing with each other. They compete even when they are just talking: They shout orders, they interrupt, they make threats and boasts, and they exchange insults. Their play is often rough and aggressive and almost always competitive. When a group of elementary-school boys play a game, sooner or later they will have a dispute about the rules; invariably, they work out a compromise and continue playing, but they may continue screaming “you cheater!” or “you liar!” while they play. Still, when the game is over, they almost always part as friends.

When boys grow up, do they change their ways of interacting with one another? Not entirely. Deborah Tannen (1990) reports one episode at a college basketball game: At the University of Michigan, student tickets have seat numbers on them, but students generally ignore those assignments and take seats on a first-come, first-served basis. One night, several men from the visiting team, Michigan State, tried to go to the seats listed on their tickets, only to find some University of Michigan students, both men and women, already seated there. The Michigan State students

FIGURE 10.37 One girl tested alone behaves about the same as one boy tested alone. But when boys play together, they “show off” to one another and to other observers; girls play more cooperatively.



asked the others to get out of their seats; the men in those seats then replied rudely, and the dispute quickly grew loud, heated, and insulting. The women were mortified with embarrassment.

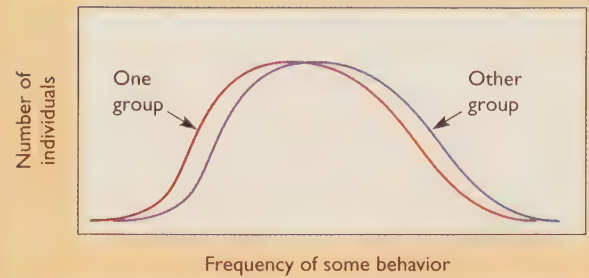
Within a few minutes, however, the Michigan State men settled into seats next to the University of Michigan students, and before long the two groups of men were happily discussing basketball strategies. The women didn't understand why the men had screamed insults in the first place or why they had made friends so quickly afterward.

Male-Female Relationships in Childhood and Adulthood What do you suppose happens when boys and girls play together? If they are working on a task that requires cooperation, few sex differences are evident (Powlishta & Maccoby, 1990). However, in an unsupervised situation with no need to cooperate, the boys often dominate and intimidate the girls. In some cases, the boys take control and the girls simply watch (Maccoby, 1990).

When boys and girls become young men and women, romantic interests may draw them together, but both are ill prepared to deal with the other sex. Men are used to demanding what they want; women are used to a cooperative give-and-take. Men worry about their status in relation to other men; women often fail to understand these status contests. When women discuss their problems, they expect their listeners to express sympathy; men often fail to understand this need. Here are some examples of the resultant misunderstanding (Tannen, 1990):

- A man invites an out-of-town friend to spend the night in the guest bedroom. The man's wife is upset that her husband did not check with her before inviting his friend. He replies that he would feel embarrassed to say, "I have to ask my wife first."
- A woman asks her husband to get their VCR to record a TV movie. The husband says this particular kind of VCR plays tapes but cannot record. The woman then asks their next-door neighbor to check the VCR. He too tells her the VCR cannot record. The husband feels resentful about this episode for years, because his wife implied that he was too incompetent to understand their VCR.
- A woman who has had breast surgery tells her husband she is unhappy about the scar left by the surgery. Instead of expressing sympathy, he replies, "You can have plastic surgery. . . ." She is upset by the implication that he doesn't like the way she looks. He replies that he doesn't care about the scar; he was trying to help because *she* said she was unhappy.

Are male-female relationships always like this? Of course not. There are many differences *on the average* between men and women, just as there are between 25-year-olds and 20-year-olds or between Northerners and Southerners, or between almost any other identifiable groups. The distributions may overlap completely; therefore, the discussion of averages may not tell us what to expect from a given individual.



Ethnic and Cultural Influences

Membership in a minority group molds a child's development in two major ways: First, the customs of the minority group may in fact differ from those of other groups. For example, many Japanese parents put great emphasis on educational achievement and bringing honor to one's family (Yamamoto & Kubota, 1983). Secondly, members of a minority group are affected by the attitudes of other people, who may treat them differently or expect certain behaviors from them simply because they are members of that minority group.

Immigrants to the United States or any other country undergo a period of acculturation, a transition from feeling part of the culture of their original country to the culture of the country they enter. Acculturation is gradual; people sometimes require a generation or more before they or their descendants feel fully comfortable in a new culture.

In many cases, a person may have to function as a member of two or more cultures or subcultures. For example,



Many immigrants are bicultural, having reasonable familiarity with two sets of customs. These Asian immigrants to Australia adopt some Australian customs while maintaining some of their own.

people who settle in an Italian-American community may speak Italian and follow Italian customs in their home neighborhood but speak English and follow U.S. customs elsewhere. African-Americans and Asian-Americans may identify strongly with their ethnic group, live in a neighborhood populated mostly by others of the same group, and rely on the help of an extended family, while also being part of the “melting pot” U.S. culture at school, on the job, and in other settings. To at least a small extent, all of us learn to function in a variety of subcultures. For example, you enter slightly different subcultures at home, at college, on the job, at a religious ceremony, and so forth. You learn to adjust what you say and do depending on the setting and the people present. Although everyone has this experience to some extent, the transitions are more noticeable and more intense for ethnic minority members.

People react to these transitions in different ways. Some minority group members become more or less completely assimilated into the majority or melting-pot culture. However, they may pay a price for doing so. For example, some high-achieving, African-American students worry that their school and occupational success may alienate them from other African-Americans—that “if you succeed, you’re betraying your color” (Fordham, 1988).

One alternative to full assimilation is **biculturalism**, the ability to alternate between membership in one culture and membership in another. Although doing so is difficult, many people succeed. The advantages of biculturalism are similar to those of bilingualism: A bilingual person (one who speaks two languages) can communicate with more people and read a greater variety of literature than can a monolingual person. Also, bilingual people come to understand their primary language from a new perspective and often become more sensitive to the multiple meanings of a word. Similarly, a bicultural person can speak with and interact with a great variety of other people and can become more aware of both the strengths and the weaknesses of each culture (Harrison, Wilson, Pine, Chan, & Buriel, 1990; LaFromboise, Coleman, & Gerton, 1993).

THE MESSAGE

Understanding and Accepting Many Ways of Life

Each of us can easily fall into the trap of thinking that our own way of growing up and relating to other people is the “right” or “normal” way. In fact, people differ substantially in their social development; we have examined some of the major reasons for this difference—temperament, family influences, gender, and ethnic and cultural identity. As a society, we are beginning to recognize and appreciate the resulting diversity of behavior that these influences create.

SUMMARY

- ★ **Temperament.** Even infants who are only a few months old show clear differences in temperament, their own characteristic way of reacting to new experiences and new people. Temperament remains fairly consistent as a person grows older. (page 382)
- ★ **Changes in the U.S. family.** As U.S. society has changed over the decades, the role of women has changed, and therefore family life has also changed. The research suggests that children who spend much of their early life in a day-care arrangement can develop without difficulties, provided that the day care is of good quality. Most children in nontraditional families develop normally, except for the effects of the poverty that is common among unmarried mothers. (page 383)
- ★ **Effects of divorce.** Children of divorcing families often show signs of distress, sometimes even before the divorce. The distress is generally more marked in European-American families than in African-American families. (page 384)
- ★ **Male-female differences.** Behavioral differences between males and females are small, on the average, when people are tested one at a time. However, in social settings males tend to associate with other males while females associate with females. Males tend to be more competitive, sometimes aggressively so. (page 385)
- ★ **Ethnic and cultural differences.** People also differ because of ethnic and cultural influences. Acculturation is the process of transition from one culture to another. Many people can function successfully as members of two or more cultures simultaneously. (page 387)

Suggestions for Further Reading

- Hetherington, E. M. (1989). Coping with family transitions: Winners, losers, and survivors. *Child Development*, 60, 1–14. Review of the effects of divorce on children.
- Maccoby, E. E. (1990). Gender and relationships. *American Psychologist*, 45, 513–520. Review of findings concerning sex differences in social behavior.
- Tannen, D. (1990). *You just don't understand*. New York: William Morrow. A popular book that discusses the various reasons that men and women often fail to understand one another.

Terms

- temperament** people’s tendency to be either active or inactive, outgoing or reserved (page 382)
- acculturation** a transition from feeling part of the culture of one’s original country to the culture of the country that one enters (page 387)
- biculturalism** the ability to alternate between membership in one culture and membership in another (page 388)

Answer to Concept Check

8. Kagan and Snidman’s study was longitudinal; they studied the same children at different ages. (page 382)

Web Resources

American Demographics

www.demographics.com/

American Demographics Magazine uses U.S. Census Bureau statistics to describe consumer trends for business leaders. Search the archives, using such search terms as *divorce*, *diversity*, *personality*, *gender*, or whatever else is of interest to you to find articles about trends in these topics.

Motivation



11

★ MODULE 11.1

General Principles of Motivation

General Properties of Motivated Behavior

Views of Motivation

Motivation as an Energy / Drive Theories / Homeostasis / Incentive Theories / Intrinsic and Extrinsic Motivations

Types of Motivation

Primary and Secondary Motivation / Maslow's Hierarchy of Needs

★ MODULE 11.2

Motivation and Biological States:

The Case of Hunger

Motives in Food Selection

Acceptance or Rejection of Foods Based on Taste / Preference for Familiar Foods / Learned Associations with Food

The Physiological Mechanisms of Hunger

Short-Term Regulation of Hunger / Long-Term Regulation of Hunger / The Brain Mechanisms of Hunger and Satiety

Eating Disorders

Obesity / The Limited Role of Emotional Disturbances / Genetics / Decreased Energy Output / Other Physiological Factors / Losing Weight / The Effect of Intentional Weight Loss on Appetite / Anorexia Nervosa / Bulimia

★ MODULE 11.3

Sexual Motivation

What Do People Do, and How Often Do They Do It?

The Kinsey Survey / Contemporary Surveys / Sexual Behavior in the Era of AIDS

Sexual Arousal

Sexual Identity and Orientation

Influences on Sexual Anatomy / Influences on Sexual Orientation / THE RELATIONSHIP BETWEEN SEXUAL

ORIENTATION AND BRAIN ANATOMY: WHAT'S THE EVIDENCE?

★ MODULE 11.4

Achievement Motivation

Measuring the Need for Achievement

The Need for Achievement and Setting Goals / Effective and Ineffective Goals

Age and Sex Differences in the Need for Achievement

The Development of Need for Achievement in Childhood / Sex Differences in Need for Achievement

Jobs That Encourage or Discourage Achievement

Motivation

During the summer of 1996, the proprietors of London's Kew Gardens announced that they had a most unusual plant, native to Sumatra and rarely cultivated elsewhere, which was about to bloom for the first time since 1963. So, if you had been in London then, with enough time available, would you have made a point of visiting Kew Gardens to witness this rare, once-or-twice-in-a-lifetime event? No? What if I told you that it was a truly beautiful flower? with a lovely, sweet smell? Still no?

Then, what if I told you the truth—that it has the nastiest, most obnoxious smell of any flower on Earth—that the name of the flower is the *stinking lily*, because it smells like a huge, week-old carcass of rotting meat or fish . . . and that one whiff of it can make a strong person retch. Now would you want to go visit the flower? If so, you would have to wait in line. When Kew Gardens announced that the stinking lily was about to bloom, an enormous crowd gathered, forming a line that stretched to the length of a soccer field (MacQuitty, 1996). (The first day that the flower bloomed, it had hardly begun to stink. Disappointed visitors vowed to return later.)

Human motivations are surprising, puzzling, and often seemingly illogical. Psychologists have made progress in understanding complex motivated behaviors, but much still remains to be learned. We begin this chapter with an overview of some general principles of motivation. Then, we shall explore three examples of motivated behaviors: hunger, sexual activity, and striving for achievement.

These examples of motivation deserve emphasis largely because they are each an important part of human life, but

also because they illustrate how our biology interacts with our social setting. Hunger is based on a biological need, but what, when, and how much we eat also depends on what we learn from other people. Sexual motivation also serves a biological need, but the search for a suitable partner is a complex social behavior. Striving for achievement is learned as a method of pleasing and impressing others. Although it is primarily a social motivation, it is also an outgrowth of the competition for dominance that we can observe throughout the animal kingdom.



There are good reasons why this flower is seldom cultivated outside its native Sumatra. Would you stand in line to visit it?

General Principles of Motivation

What is motivation?

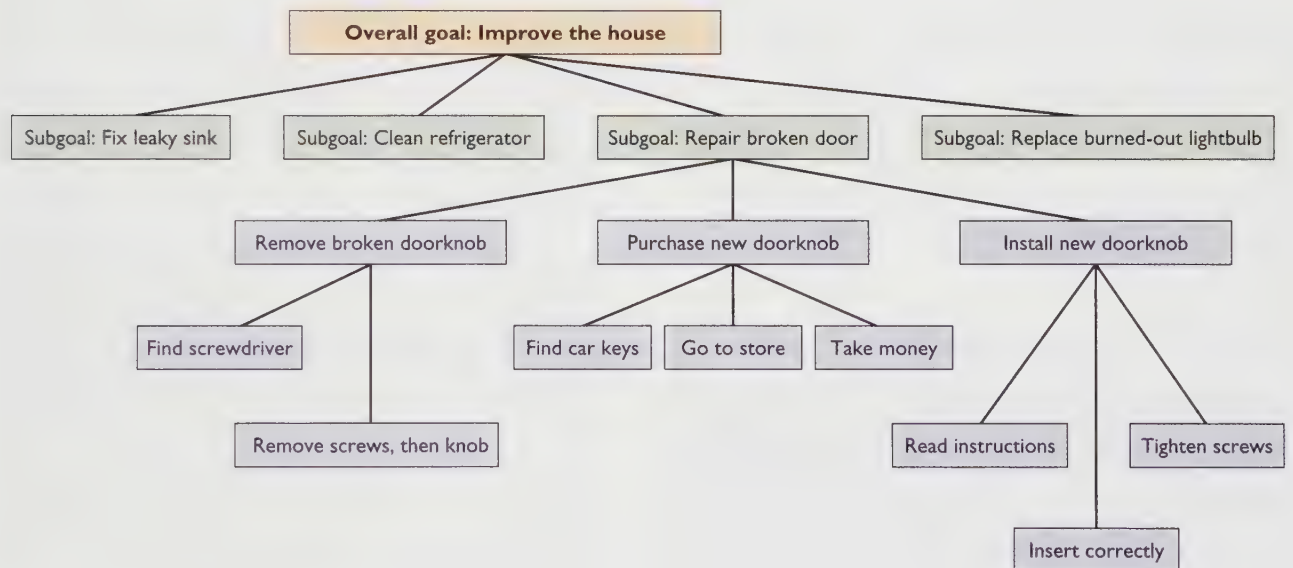
How could a psychologist determine whether an act is motivated?

You are sitting quietly, reading a book, when suddenly you hear a loud noise. You jump a little and gasp. Was that

was an accident." Your friend, who promised to drive you somewhere and then left without you, says, "I didn't do it on purpose. I just forgot." Maybe so and maybe not. How can we determine whether a behavior is motivated or accidental, intentional or unintentional? We need a clear understanding of how motivated behaviors differ from unmotivated behaviors.

General Properties of Motivated Behavior

What, if anything, do the various types of motivated behavior have in common? The foremost characteristic of motivated behaviors is that they are goal directed. Highly motivated individuals persist with a behavior or alter their behavior until they reach their goal; they frequently set up subgoals that they will need to achieve on the way to their final goal (Austin & Vancouver, 1996). For example, if you were motivated to improve your house, you might make a list of the items that need repair and then determine the tools needed for each repair:



action motivated? "No," you say. "I jumped involuntarily." Now I tell you that I want to do a little experiment. I shall tap my pencil; as soon as you hear it, you should try to jump and gasp just as you did the first time. I tap my pencil and, sure enough, you jump and gasp. Was that action motivated? "Yes," you reply.

So, what appears to be approximately the same behavior can be motivated at one time and unmotivated at another time. Can we trust people to tell us whether their behavior was motivated or unmotivated? Not always. Someone accused of murder says, "I didn't mean to kill him. It

To determine whether a particular behavior is motivated, as opposed to automatic or reflexive, an observer needs to watch an individual over a period of time in a variety of circumstances. If the individual varies the behavior at different times and persists until reaching a goal, then the behavior is motivated, or intentional.

SOMETHING TO THINK ABOUT

A frog flicks its tongue at a passing insect, captures it, and swallows it. The behavior serves to satisfy the frog's need for

food, so we might guess that it is motivated. However, the behavior appears to be as constant as a reflex. How might you determine whether this behavior is motivated? *

A second characteristic of motivated behaviors is that they vary from time to time and from one individual to another, under the influence of both internal (biological) and external (social) controls. For example, you wear different clothes from one day to the next, and you wear different clothes from your roommate on any given day.

Views of Motivation

What is motivation? Let's try some definitions: "Motivation is what activates and directs behavior." That description fits many examples well enough, but it also fits other, nonmotivational phenomena. For example, light activates and directs the growth of plants, but we would hardly say that light motivates plants.

"Motivation is what makes our behavior more vigorous and energetic." The problem with that definition is that some people are strongly motivated to lie motionless for hours on end.

How about this: "Motivation is what changes one's preferences or choices"? That might do, except that we would first have to define preference and choice.

Frankly, it is hard to state precisely what we mean by motivation. Psychologists have repeatedly altered their views of motivation. By considering one theory after another, they have seen the shortcomings of each and have in the process developed some idea of what motivation is and is not.

Motivation as an Energy

Motivation, which comes from the same root as *motion*, is literally something that "moves" a person. So, we might think of it as a type of energy. According to Konrad Lorenz (1950), a pioneer in the study of animal behavior, animals engage in instinctive acts when specific energies reach a critical level. For example, a male stickleback fish has no specific energy for mating outside the breeding season, so it will not respond sexually. At the start of the breeding season, it has a small amount of mating energy, and it will then court female stickleback fish and attack male stickleback fish. At the height of the breeding season, it has a great amount of mating energy, so it will court females vigorously; it may even respond sexually to a piece of wood painted to resemble a female of its species.

Figure 11.1 illustrates Lorenz's model. A specific kind of energy builds up in the reservoir and flows into the tray below. The outlets in that tray represent ways of releasing this energy. If conditions are right, the energy is released through the lowest outlet—for example, mating with a

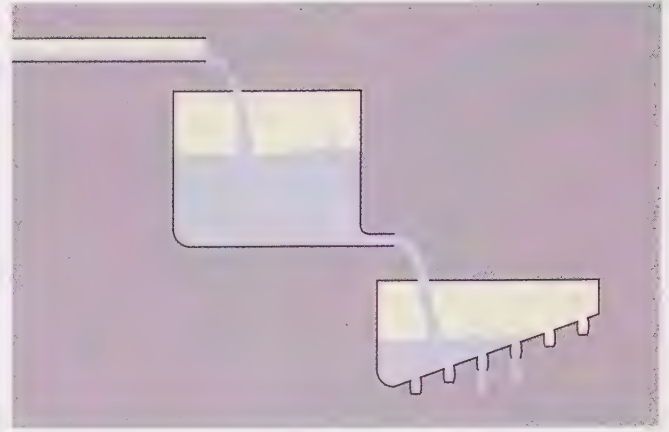


FIGURE 11.1 According to Konrad Lorenz, energy (represented as a fluid) builds up in a "reservoir" in the brain and needs to be discharged. Ordinarily, an instinct is released through natural or preferred outlets. If those outlets are blocked, however, energy spills over into another outlet, and the animal engages in an unrelated behavior. (After Lorenz, 1950.)

normal partner. If that outlet is blocked and energy continues to build up, the energy will spill through one of the higher, less preferred outlets. However, Lorenz's theories reflect an obsolete conception of how the nervous system works. He believed that every impulse to action had to be carried out, so that if one response was blocked, the energy would activate something else. We now know that an individual can simply inhibit the impulses toward disadvantageous behavior.

Drive Theories

Related to the instinctive energy theories are theories that describe motivation as a **drive**, *a state of unrest or irritation that energizes one behavior after another until one of them removes the irritation* (Hull, 1943). For example, when you get a splinter in your finger, the discomfort motivates you to engage in various actions until you get rid of the splinter.

According to the *drive-reduction theory* that was popular among psychologists of an earlier era, animals including humans strive to reduce their needs and drives as much as possible. That is, they eat to reduce their hunger, drink to reduce their thirst, have sexual activity to reduce their sex drive, and so forth. This view implies that we all strive for a state where all our needs have been met, at which time we would become inactive. The primary shortcoming of this view is that, in fact, most people seek variety and activity in life; the ideal state is one with a moderate amount of stimulation, not one with as little as possible.

Another flaw in this drive theory is that it ignores the role of external stimulation. For example, a person's interest in food depends not only on hunger (an internal drive) but also on what foods are available. Similarly, interest in sex depends partly on an internal drive and partly on the presence or absence of a suitable partner.

Homeostasis

One important advance upon the idea of drive reduction is the concept of **homeostasis**, *the maintenance of an optimum level of biological conditions within an organism* (Cannon, 1929). The idea of homeostasis recognizes that we are motivated to seek a state of equilibrium, not to reduce all stimuli to zero. For example, people maintain a nearly constant body temperature through a combination of physiological and behavioral means. We fight against both increases and decreases in body temperature. Each of us also works to maintain a fairly steady body weight, a nearly constant amount of water in the body, a reasonably stable level of sensory stimulation, and so on.

Unlike a rock, which remains static only because nothing is acting upon it, the homeostasis of the body is more like a spinning top; someone must apply additional energy from time to time to keep it spinning. For example, we maintain constant body temperature partly by shivering, sweating, and other involuntary physiological responses and partly by putting on extra clothing, taking off excess clothing, or finding a more comfortable location.

Human motivated behaviors differ from the actions of a thermostat in one important regard: Our behavior often anticipates future needs. For example, you might eat a large breakfast one morning, even though you are not terribly hungry, just because you know you are going to be too busy to stop for lunch. If you are angry or frightened, you begin to sweat even before you begin the vigorous actions that might heat up your body. (We call this phenomenon a “cold sweat.”) Thus, one fruitful way of describing motivation is that it maintains current homeostasis and anticipates future needs to maintain future homeostasis (Appley, 1991).

Still, even that conception of motivation overlooks the power of new stimuli to arouse motivated behaviors. For example, nonhungry people may eat or drink just to be sociable or because someone has offered them something especially tasty.

Incentive Theories

Why do people ride roller coasters? It is doubtful that they have any special need to go thundering down a steep decline. Or suppose you have just finished a big meal and someone offers you a slice of a very special cake. If you are like most people, you eat it but not because you need it. Evidently, motivation includes more than the internal forces that push us toward certain behaviors; it also includes **incentives**—*external stimuli that pull us toward certain actions*.

The distinction between a drive and an incentive is not clear-cut. Jumping into a swimming pool on a hot summer day may satisfy your biological drive to maintain normal body temperature, but the prospect of splashing around in the water may serve as a strong incentive as well.

Most motivated behaviors are controlled by a combination of drives and incentives. You eat because you are hungry (a drive) and because you see appealing food in front of you (an incentive). How much you eat depends on both the strength of the drive and the appeal of the incentive.

Intrinsic and Extrinsic Motivations

Similar to the distinction between drives and incentives, psychologists also distinguish between intrinsic and extrinsic motivations. An **intrinsic motivation** is *a motivation to engage in an act for its own sake*; an **extrinsic motivation** is *based on the reinforcements and punishments that the act may bring*. For example, if you eat because you are hungry, you are following an intrinsic motivation; if you eat something you don't like just to please the cook, you are following an extrinsic motivation. Most behaviors follow a combination of both kinds of motivation. For instance, an artist paints partly for the joy of creation (intrinsic) and partly for the eventual profit (extrinsic). You read this book partly because you enjoy reading it (I hope), partly because you want to get a good grade in the course, partly because you know you will feel guilty if you spend a lot of money on a college education and then fail your courses, and partly because you saw your roommate studying a textbook and decided to follow the example.

Does a combination of intrinsic and extrinsic motivations lead to more persistent and effective performance than, say, an intrinsic motivation alone? Not always. In a classic study, researchers let four monkeys play with a device like the one in Figure 11.2. To open it, a monkey had to remove the pin, lift the hook, and then lift the hasp, in that order. The monkeys played with the device from time to time over a period of 10 days. They received no reinforcements; they played with it apparently just for the fun of it (an intrinsic motivation). By the end of the 10 days, each



We eat because of both intrinsic and extrinsic motivations. Even when the hunger (an intrinsic motivation) is satisfied, we eat because of the taste and the desire to socialize (extrinsic motivations).

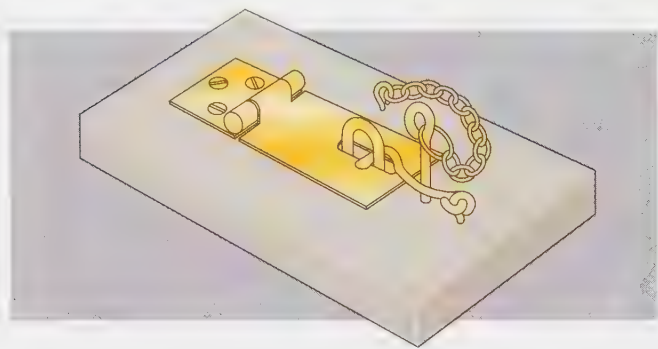


FIGURE 11.2 Monkeys learned to open this device by removing the pin, the hook, and the hasp, in that order. At first, they received no reward but continued to open the device just for the fun of it. Then, when they could open it to obtain a raisin, their performance deteriorated. Evidently, in some cases an individual performs better with just intrinsic motivation (here, the joy of the task itself) than with a combination of intrinsic and extrinsic motivations.

monkey was able to open the device quickly, almost always taking the steps in the right order. Then, the device was placed over a food well in a place where the monkeys were accustomed to finding a raisin (an extrinsic motivation). If they opened the device they could get the raisin. Suddenly, their ability to open the device deteriorated. Instead of patiently removing the pin, the hook, and the hasp as they had done before, they attacked the hasp forcefully. They took longer to open the device for food than they had for play. Later, when they were offered the device by itself with no food available, they opened it less frequently than before and made more errors in their attempts (Harlow, Harlow, & Meyer, 1950). Evidently, opening the device for food had become work, so the monkeys no longer saw it as play.

Might the same principle apply to human behavior? In a typical experiment to test this idea, college students were asked to try to arrange seven plastic pieces with complex shapes to match figures in a drawing. At one point halfway through the experiment, students in the experimental group were paid \$1 for each correct match. (Students in the control group did not know that the experimental group was being paid.) Then, the experiment continued without pay for anyone. After pay was suspended, the experimental group did less work than the control group did (Deci, 1971). Results such as these illustrate the **overjustification effect**: *When people are given more extrinsic motivation than necessary to perform a task, their intrinsic motivation declines.* According to one interpretation, people ask themselves, “Why am I doing this?” They then answer, “It’s not because I enjoy the task. It’s because I’m being paid.” If they stop getting paid, they lose interest in the task.

The overjustification effect, like various other psychological principles, applies under some conditions but not others (Eisenberger & Cameron, 1996). It is wrong to conclude, as some have, that we should never reward people or even praise them for a job well done. Verbal praise, especially if it is well earned, strengthens people’s enjoyment of a task. The overjustification effect applies only when people are given physical rewards for doing a task, and even then it is sometimes a weak effect.

Table 11.1 summarizes four views of motivation.

CONCEPT CHECK

1. Suppose you want to encourage your younger cousin to continue taking piano lessons. Based on the overjustification effect, would it be wise to pay him for practicing? (Check your answer on page 399.)

TABLE 11.1 Four Views of Motivation

VIEW	BASIC POSITION	MAJOR WEAKNESSES
Instinct theories <i>According to instinct theories, motivation is a kind of energy that builds up until it finds a release.</i>	Motivations are energies that accumulate; each energy specifies a preferred action, although it might spill over into a less preferred outlet.	Based on obsolete view of the nervous system.
Drive theories <i>According to drive theories, motivation is an irritation that continues until we find a way to reduce it.</i>	Motivations are based on needs or irritations that we try to reduce; they do not specify particular actions.	Implies that we always try to reduce stimulation, never to increase it. Also overlooks importance of external stimuli.
Homeostasis (plus anticipation) <i>Homeostasis is the process of maintaining a variable, such as body temperature, within a set range.</i>	Motivations tend to maintain body states near some optimum, intermediate level. They may react to current needs and anticipate future needs.	Overlooks importance of external stimuli.
Incentive theories <i>Incentives are external stimuli that attract us, even if we have no biological need for them.</i>	Motivations are responses to attractive stimuli.	Incomplete theory unless combined with drive or homeostasis.

Types of Motivation

How many motivations do people have? They are motivated to obtain food, water, shelter, and clothing; to have social contact with others; to smell the stinking lily in Kew Gardens. . . . The list could go on and on. Can we group these motivations into a few coherent categories?

Primary and Secondary Motivation

One way to categorize motivations is to distinguish primary motivations from secondary motivations. **Primary motivations**—such as the search for food and water—are *automatic, built-in processes*. **Secondary motivations develop as a result of specific learning experiences**, presumably because the secondary motivation has, in the past, led to the satisfaction of a primary motivation. Primary motivations and secondary motivations are analogous to the unconditioned reinforcers and conditioned reinforcers that we considered in Chapter 6.

Presumably, we learn secondary motivations because they help us to satisfy primary motivations. For example, we learn to have a desire for money (a secondary motivation) because it helps us to obtain food, water, and shelter (primary motivations). Often, however, a secondary motivation seems to develop a momentum of its own, thus becoming apparently independent of the original primary motivations associated with it. For example, many people start a coin or stamp collection because they think it might be worth something and eventually start spending money to add to the collection for its own sake. Some people try to do something spectacular to get into the *Guinness Book of World Records*. Why? to achieve fame? In any case, they sometimes lose



This man is trying to set the world record for being covered by bees. The desire for fame is an amazingly strong motivator.

sight of the fact that they are devoting more effort than the fame or any other reward is likely to be worth. One man holds the record for the longest fingernails, at 41.6 inches and still growing; another man has been growing his fingernails for 30 years in hopes of eventually catching up and surpassing the current champion. Other people have performed such feats as smoking 114 cigarettes in 3 minutes, pushing a pea with their nose for 3 miles, French-kissing snakes, and allowing their body to be covered with a record number of scorpions—all in failed attempts to get into the *Guinness* book (Spaeth, 1995). (The Guinness people refuse to publicize such records because they don't want to encourage self-destructive behaviors.)

CONCEPT CHECK

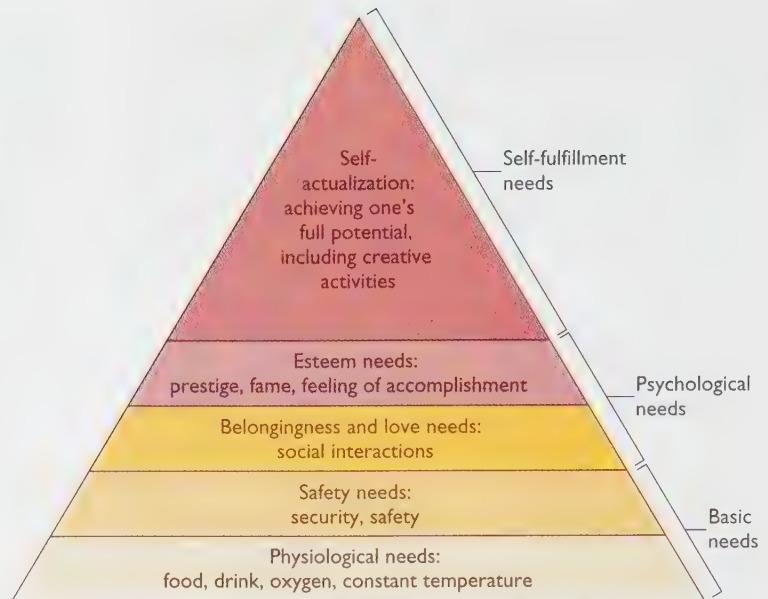
2. Is your interest in graduating from college a primary motivation or a secondary motivation? (Check your answer on page 399.)

Maslow's Hierarchy of Needs

Abraham Maslow (1970) attempted to bring some organization to the listing of human motivations, including both primary and secondary motivations. According to Maslow, our behavior is governed by a **hierarchy of needs, an organization from the most necessary and insistent to the ones that receive attention only when all others are under control**. The most basic are the physiological needs for food, drink, oxygen, and warmth, as shown at the bottom level of Figure 11.3. According to Maslow, these basic needs ordinarily take priority over all others (Figure 11.4). For example, people who are gasping for breath will not take time out to do something else until they have satisfied their need for oxygen. Once people have satisfied all of their physiological needs, they seek to satisfy their safety needs, such as security from attack and avoidance of pain. When those needs are satisfied, they proceed to the needs for love and belonging—making friends and socializing with them. Next come the needs for esteem, such as gaining prestige and a feeling of accomplishment. At the apex of Maslow's hierarchy is the need for **self-actualization, the need to achieve one's full potential**.

Maslow's theory is appealing because it recognizes that the various motivations are not equal. When they conflict, people will generally give priority to the tasks most necessary for survival. However, there is no evidence to support the idea that motivations fall into five distinct categories (Wahba & Bridwell, 1976). That is, the differences between the need for oxygen and the need for food (both basic physiological needs) are as great as the differences between the need for love and the need for self-esteem. Furthermore, people sometimes work to satisfy higher-level needs before they satisfy lower-level needs. Even when you are ravenously hungry, you might skip a meal to be with someone you love, or to study for a test, or to accept an award. Martyrs have willingly

FIGURE 11.3 Maslow's hierarchy of needs suggests a hierarchical order to human motivations. If you are thirsty, you will want to drink something to fill that basic need. If your primary physical needs are met, you can then focus on meeting your psychological needs for companionship and achievement. Your final efforts will be devoted to reaching your full potential. But do all people follow this principle? Some people, including Olympic performers, sacrifice eating, entertainment, and companionship to strive toward athletic accomplishment and prestige. Artists might also give up physical comforts and social pleasures to focus on their inner vision and to realize their creative potential.



sacrificed their lives to advance a political or religious cause. Depending on the circumstances, almost any motivation may take priority over the others, at least temporarily.

THE MESSAGE

Many Types of Motivation

People frequently puzzle over why someone did something. For example, why would someone stand in line for hours to smell a stinking lily? We almost exclusively puzzle over behaviors that contribute little or nothing to survival. That



FIGURE 11.4 According to Maslow's hierarchy of needs, we concentrate first on the lowest level of needs—such as the need for food, oxygen, and constant temperature—until we meet them. Then we move on to higher and higher needs. Thus, an impoverished and homeless person would be unlikely to devote much effort to creative endeavors. (Some exceptions to this rule do occur, however.)

is, we do not bother to ask why when someone gasps for breath, eats, drinks, or runs away from a tiger. For the unusual behaviors such as trying to smell a stinking lily, people seldom have just one simple motivation. More often, it is a combination: curiosity, the desire to have something unusual to talk about with friends, perhaps even a desire for fame of a very local variety.

S U M M A R Y

- ★ **Characteristics of motivated behaviors.** Motivated behaviors persist until the individual reaches the goal. They are controlled by internal and external forces and by biological and social forces. Motivated behaviors vary from time to time, from situation to situation, and from person to person. (page 393)
- ★ **Motivation as energy or drive reduction.** Psychologists have sometimes viewed motivation as an energy that must be used up in one way or another, or as a process that persists until all drives are reduced. Each of these views has serious limitations. (page 394)
- ★ **Motivation as a way of maintaining homeostasis.** To a large degree, motivated behaviors tend to maintain body conditions and stimulation at a near-constant, or homeostatic, level. This view of motivation can account for much behavior, if we also assume that behaviors anticipate future needs instead of just responding to current needs. However, the homeostatic view of motivation overlooks the role of external stimuli for arousing behavior. (page 395)
- ★ **Motivation as incentive.** Motivations are partly under the control of incentives—external stimuli that pull us toward certain actions. Both drives and incentives control most motivated behaviors. (page 395)
- ★ **Intrinsic and extrinsic motivations.** People and animals engage in some actions because the actions themselves

are interesting or pleasing (intrinsic motivation). Providing a physical reinforcement (extrinsic motivation) for the actions sometimes backfires by reducing the interest or pleasure they provide. (page 395)

★ **Types of motivations.** Psychologists have made several attempts to list or categorize various motivations. One prominent attempt, offered by Abraham Maslow, arranged needs in a hierarchy ranging from basic physiological needs at the bottom to the need for self-actualization at the top. His claim that people satisfy their lower needs before their higher needs does not apply in all cases, however. (page 397)

Terms

drive an internal state of unrest or irritation that energizes one behavior after another until one of them removes the irritation (page 394)

homeostasis the maintenance of biological conditions at an optimum level within an organism (page 395)

incentive an external stimulus that prompts an action to obtain the stimulus (page 395)

intrinsic motivation motivation to engage in an act for its own sake (page 395)

extrinsic motivation motivation based on the rewards and punishments that an act may bring (page 395)

overjustification effect the tendency of people who are given more extrinsic motivation than necessary to perform a task to experience a decline in their intrinsic motivation (page 396)

primary motivation motivation that serves biological needs (page 397)

secondary motivation motivation that serves no direct biological need but develops as a result of specific learning experiences (page 397)

hierarchy of needs Maslow's categorization of human motivations, ranging from basic physiological needs at the bottom to the need for self-actualization at the top (page 397)

self-actualization the need to achieve one's full potential (page 397)

Answers to Concept Checks

1. According to the overjustification effect, you should not pay him enough that he starts practicing just for the reward. However, verbal praise would be good. (page 396)
2. Your interest in graduating from college is a secondary motivation, because it is something you had to learn to value. Such secondary motivations can become very strong. (page 397)

Web Resources

Autobiography of Konrad Lorenz

www.nobel.se/laureates/medicine-1973-2-autobio.html

Konrad Lorenz is one of psychology's few Nobel Laureates. The 1973 Nobel Prize in Physiology or Medicine was jointly awarded to Konrad Lorenz and two other ethologists "for their discoveries concerning organization and elicitation of individual and social behaviour patterns." The press release (www.nobel.se/laureates/medicine-1973-press.html) explains the significance of their contributions.

Maslow's Hierarchy of Needs Theory

snycorva.cortland.edu/~andersmd/maslow/homepage.html

A brief tutorial about Maslow's theory of motivation, the hierarchy of needs, and the application of the theory to education. Can you spot the fairly major error in the description of the hierarchy?

Motivation and Biological States: The Case of Hunger

How do we choose which foods to eat?

Why do some people gain so much more weight than others do?

Small birds eat only as much as they need to meet their immediate needs, storing almost no fat at all. The advantage is that they remain light and therefore fast, just as they need to be to escape predators. The disadvantage is that they have no reserves; if they have trouble finding food for even a short time, they can starve to death. Bears follow a different strategy. Sometimes they can find food in abundance and sometimes they find almost nothing for days or even weeks at a time. So, they feast on all the food they can eat, when they can find it, and store enough fat to meet their energy needs during times of famine.

Which strategy is better for humans? It depends. Something midway between these two extremes is probably best, although the ideal varies depending on circumstances:



FIGURE 11.5 Mealtime is more than just an opportunity to satisfy hunger: It is an occasion to bring the family and sometimes friends together, to share a pleasant experience, to discuss the events of the day, and even to pass on a culture from one generation to the next. We expect family members to participate in these meals, even if they are not hungry.

Can you always find good food, or might there be a famine? Does your survival depend on being as quick as a small bird, or could you afford to slow down like a fat bear and still survive?

How much we eat, and what and when we eat, also depends on factors that have nothing to do with our energy needs (Figure 11.5). When you visit friends or relatives, they may offer you food as a gesture of affection, and they may act hurt if you refuse their hospitality. Say that you visit the family of your boyfriend or girlfriend, and you want to make a good impression. “Dinner’s ready!” someone calls. You go into the dining room and find a huge meal spread out before you, which your hosts clearly expect you to enjoy. Do you explain that you are not hungry because you already made a pig of yourself at lunch? Probably not.

Motives in Food Selection

Have you ever wondered how people first figured out which foods were edible? It was easy for you and me; our parents and other people told us what to eat, and we did not have to try all the weeds in the field to decide which ones were good. Our parents learned from their parents and they in turn from their parents, but way back when, somebody had to discover what was edible by trial and error. Furthermore, they had to discover many principles of how to prepare foods (Rozin, 1996). For example, cassava, a root native to South America, is poisonous unless someone washes it and pounds it for about 3 days. Cashews are covered with a thin membrane that must be carefully removed; anyone who touches it will react as if they had touched poison ivy. American corn (maize) has an abundance of certain nutrients and a deficit of others, and beans have nearly the opposite set of abundances and deficits, so it is best to combine corn with beans—as the Native Americans discovered long ago.

We learn our food choices largely by learning what *not* to eat. Toddlers around the age of 1½ will try to eat almost anything they can fit into their mouths (Figure 11.6). Up to the age of 7 or 8, about the only reason children give for refusing to eat something is that they think it would taste bad (Rozin, Fallon, & Augustoni-Ziskind, 1986). As children grow older, they give a wider variety of reasons for accepting certain foods and rejecting others. As with other motivations, food selection depends on a combination of physiological, social, and cognitive factors. Let’s consider some of the most important factors.

Acceptance or Rejection of Foods Based on Taste

Some taste preferences are present at birth. Infants readily consume sweet liquids; when they taste something bitter or sour, they turn their head and spit it out.



FIGURE 11.6 Infants and young children will try eating almost anything and refuse food only if it tastes bad. As they grow older, they learn to avoid foods for other reasons. People avoid eating some substances to avoid getting sick; they avoid other substances because the very idea of eating them is disgusting.

At least one taste preference can be triggered by an abnormal condition within the body. One boy showed a strong craving for salt. As an infant, he licked the salt off crackers and bacon but refused to eat the food itself. One of the first words he learned was *salt*. He put a thick layer of salt on everything he ate, and sometimes he ate salt by itself. When deprived of salt, he ate almost nothing and began to waste away. At the age of 3½, he was taken to the hospital and fed the usual hospital fare. He soon died of salt deficiency (Wilkins & Richter, 1940).

It turned out that the boy's adrenal glands were defective. These glands secrete hormones that enable the body to retain salt. The boy craved great amounts of salt because salt was being excreted so rapidly from his body. (We are often told to limit our salt intake for health reasons, but too little salt can also be dangerous.)

Research on animals confirms that a deficiency of salt in the body triggers an immediate craving for salty foods. As soon as animals, including humans, become salt deficient, they show a heightened preference for salty tastes (Rozin & Kalat, 1971). People who have lost large quantities of salt as a result of bleeding or heavy sweating often express a craving for salt. Apparently, salty foods actually taste better to salt-deficient people and animals than they do to others (Jacobs, Mark, & Scott, 1988). In short, changes in body chemistry can alter a person's motivation to choose a particular food.

Preference for Familiar Foods

Although most people eat a wide variety of foods, they are cautious about eating anything unfamiliar. Think about the first time you tried jalapeño peppers or coffee, for example. Even foods that taste reasonably good at first—artichokes, for example—taste even better after they become familiar.

Members of every culture and every ethnic group become familiar with its preferred ways of preparing and seasoning foods (Rozin, 1996). Children who grow up in Italian families come to prefer foods flavored with tomato, garlic, and olive oil. Mexican infants dislike hot peppers at first, but by the time they are a few years old, many insist on having them with almost every meal. Cuisine is one of the most stable and defining features of any culture. In one study, researchers interviewed Japanese high-school and college students who had spent a year in another country as part of an exchange program. The researchers asked the students how much they enjoyed their experience and how much distress they felt at various times. The amount of pleasure or displeasure had little relationship to the religion, family life, recreation, or dating customs of the host country. The main determinant was the food: Students who could at least occasionally get something similar to Japanese food had a good time; those who could not became unhappy and homesick (Furukawa, 1997).

Learned Associations with Food

As I mentioned in Chapter 6, animals associate foods with the gastrointestinal consequences of eating them. The same is true of humans. When you eat something and later feel sick, you may form a strong aversion to that food, especially if it was an unfamiliar one. Ordinarily, the aversion forms because something in the food made you ill, but the same learning takes place regardless of what caused the illness. If you eat something at an amusement park and then go on a wild ride and get sick, you may never again like that food (Figure 11.7). Even though you know the ride was at fault, an area deep in your brain still associates the food with the sickness.

People can also develop preferences by associating a food with another food that they already enjoy (Capaldi, 1996). For example, parents who want their child to learn to like broccoli might mix a little broccoli with some cheese or other food that the child already likes. Later, they could gradually reduce the amount of cheese.

People also reject safe, nutritious foods because of the very idea of where those foods came from (Rozin & Fallon, 1987; Rozin, Millman, & Nemeroff, 1986). In the United States, most people refuse to eat dog meat, cat meat, or horse meat. Many vegetarians consider any kind of meat disgusting and are distressed even to watch other people eat it. The longer people have been vegetarians, the more firmly they tend to reject meat-eating as not only undesirable but even immoral (Rozin, Markwith, & Stoess, 1997).

How would you like to try the tasty morsels described in Figure 11.8? Most people find the idea of eating insects



FIGURE 11.7 People associate the foods they eat, especially unfamiliar foods, with the way they feel afterward. If you ate corn dogs and cotton candy just before getting on a wild roller coaster ride and then got sick from the ride, something in your brain would “blame” the food for your feeling ill, even though you consciously believe that the food had nothing to do with your illness. Ordinarily, however, this kind of learning teaches us to avoid harmful substances.

repulsive, even if the insects were sterilized to kill all the germs (Rozin & Fallon, 1987). People also say they would refuse to drink a glass of apple juice after a dead, sterilized cockroach had been dipped into it. After seeing a cockroach dipped into a glass of apple juice, some people even refuse to drink other apple juice that was poured into a different glass (Rozin, Millman, & Nemeroff, 1986).

Crispy Cajun Crickets

Adapted from a recipe in the *Food Insects Newsletter*, March 1990

Tired of the same old snack food? Perk up your next party with Crispy Cajun Crickets (“pampered” house crickets, *Acheta domesticus*, available from Flucker’s Cricket Farm, P.O. Box 378, Baton Rouge, LA 70821, 800-735-8537).

- 1 cup crickets
- 1 pinch oatmeal
- 4 ounces butter, melted
- Salt
- Garlic
- Cayenne

1. Put crickets in a clean, airy container with oatmeal for food. After one day, discard sick crickets and freeze the rest.
2. Wash frozen crickets in warm water and spread on a cookie sheet. Roast in a 250-degree oven until crunchy.
3. Meanwhile heat butter with remaining ingredients and sprinkle this sauce on crickets before serving.

Yield: 1 serving



Different cultures have different taboos. Here is an assortment of insect and reptile dishes. (Yum, yum?)

The Physiological Mechanisms of Hunger

Hunger is a (partly) homeostatic drive that serves to keep fuel available for the body to use. Specialized mechanisms in the brain monitor how much fuel is available; when supplies begin to drop, the brain triggers behaviors that lead to eating. But how does the brain know how much fuel is available and, therefore, how much a person should eat and how often?

The problem is far more complex than keeping enough fuel in the gas tank of a car. When the fuel gauge shows that the tank is running low, you fill it with gas. By contrast, keeping track of how much fuel is in your stomach does not tell you how much more you need. Right now, in addition to the fuel in your stomach and intestines, a fair amount of fuel is present in every cell of your body, ready to be used. Additional fuel is circulating in your blood, ready to enter cells that need it. Still more fuel is stored in the fat cells, available to be converted into a form that can enter the bloodstream. If necessary, your body can break down muscle tissues to provide additional fuel. Your car will stop within seconds after it uses up all the fuel in the gas tank, whereas your body can keep going for days, even weeks, after your stomach is empty.

How do you know how much to eat, especially considering that each meal has a different density of nutrients from any other? Fortunately, you don’t have to get it exactly right. You have one set of mechanisms that control short-term changes in hunger and a separate set of long-term mechanisms that act as a correction if your short-term mechanisms cause you to eat too much or too little.

Short-Term Regulation of Hunger

The factors responsible for ending a meal are several, but they are easy to describe. Under most circumstances, the main factor is distension of the stomach and intestines; we

FIGURE 11.8 People avoid eating some potential foods because they are disgusted by the very idea of eating them. For example, most people would refuse to eat insects, regardless of any assurances that the insects were nutritious and harmless.

feel “full” because the digestive system is literally full (Seeley, Kaplan, & Grill, 1995). We also monitor how much we have eaten (Spiegel, 1973); with familiar foods we calibrate approximately how much nutrition we are getting per amount swallowed (Deutsch & Gonzalez, 1980); and we have receptors in the intestines that detect the sugars in the food (Lavin et al., 1996). The fact that these receptors detect sugars and not fats may be one reason that people tend to overeat on a high-fat diet.

The main factor responsible for the onset of hunger and the start of a meal is a drop in how much glucose enters the cells, but the control of glucose is a more complicated story (Figure 11.9). **Glucose, the most abundant sugar in your blood, is an important source of energy for all parts of the body and almost the only source for the brain.** Most foods can be converted at least partly into glucose. Excess

blood glucose can be converted into fats and other stored fuels, and stored fuels can be converted back into blood glucose when necessary. The flow of glucose to or from the blood depends on two hormones released by the pancreas, insulin and glucagon.

Insulin is a hormone that increases the flow of glucose and several other nutrients into body cells. At the beginning of a meal, before the nutrients have even started entering the blood, the brain sends messages to the pancreas to increase its secretion of insulin. Insulin promotes the movement of glucose and other nutrients out of the blood and into both the cells that need fuel (such as muscles and neurons) and the cells that store the nutrients as fats and other supplies. As the meal continues, the digested food enters the blood, but almost as fast as it enters, insulin helps to move excess nutrients out of the blood and into the liver

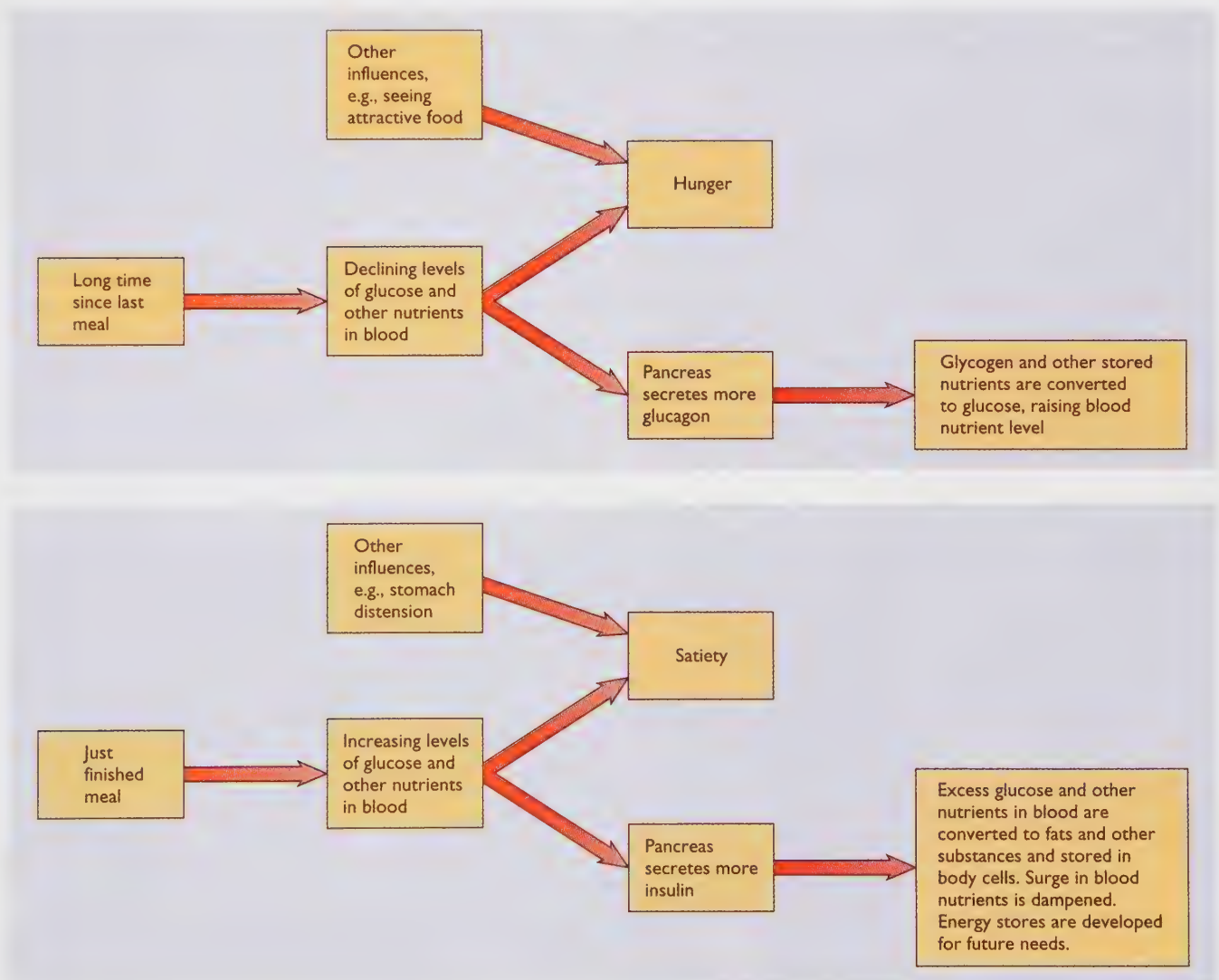


FIGURE 11.9 The short-term regulation of eating depends on the levels of glucose and other nutrients in the blood; it also depends on the appearance and flavor of the food, social influences, and so forth. Varying secretions of the hormones insulin and glucagon help to keep the blood nutrient levels reasonably constant. During and shortly after a meal, insulin moves blood nutrients into storage in the liver and fat cells; during a period without food, glucagon converts stored nutrients into blood glucose.

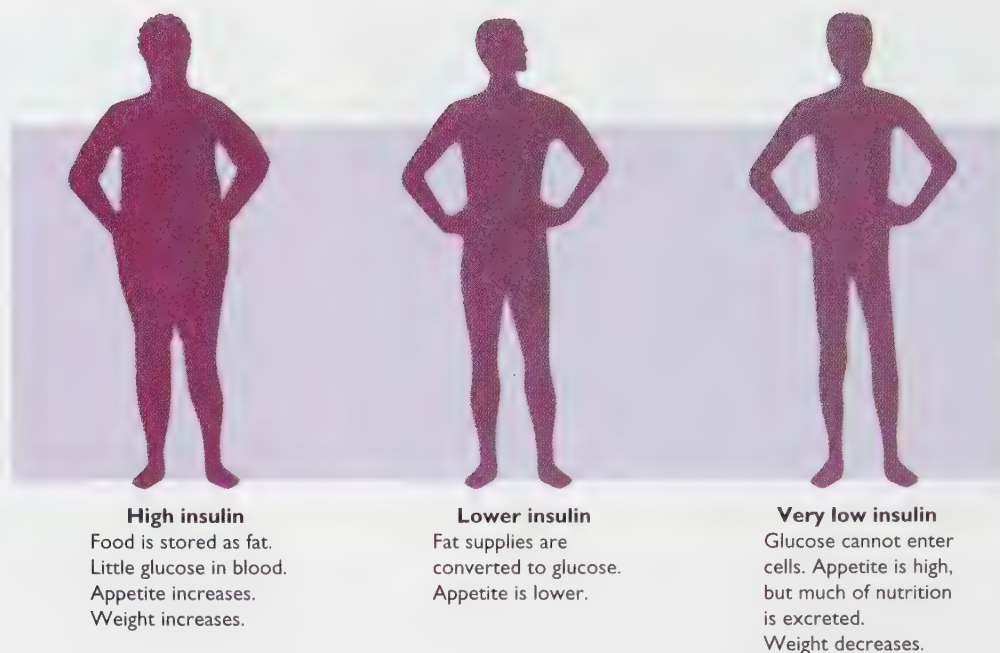


FIGURE 11.10 How insulin affects glucose, appetite, and weight.

or fat cells. In that manner, the insulin holds down the surge of glucose and other nutrients in the blood (Woods, 1991). Later, long after the meal, when the nutrient supply in the blood starts to drop, the pancreas reduces its secretion of insulin and increases its secretions of **glucagon**, a hormone that helps to convert stored energy supplies back into blood glucose.

At least that is the story for healthy people. Consider what happens if insulin levels stay constantly high or low: When insulin levels are consistently low, as with the medical condition diabetes, nutrients enter the cells very slowly (Figure 11.10). People with untreated diabetes eat without satisfying their hunger because very little of the food enters

the cells. It does not even enter the fat cells, so they can eat a great deal without gaining weight (Lindberg, Coburn, & Stricker, 1984). They simply excrete much of what they eat.

At the opposite extreme, if insulin levels are consistently high, nutrients enter the cells easily, but a large percentage of every meal is converted to fats and stored in fat cells. Because the insulin level remains high, the food stored in fat cells simply stays there; glucagon cannot mobilize much of it back into blood glucose. Consequently, soon after a meal, the person again has a low level of blood glucose (*hypoglycemia*) and an increased appetite. (Figure 11.11 shows the relationship between glucose level and food intake.) Note that, if the insulin level is either consistently low

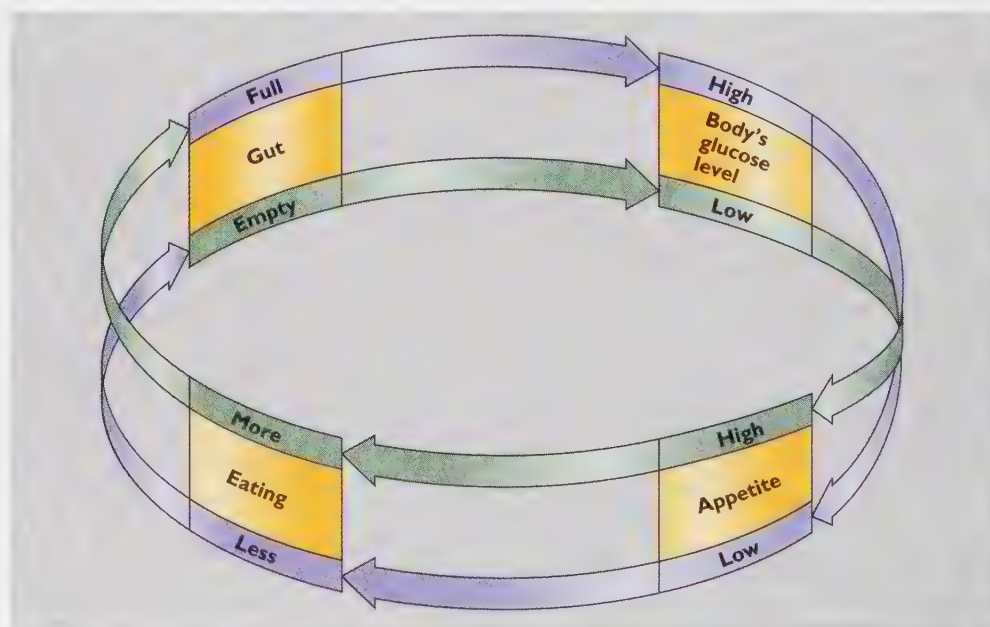


FIGURE 11.11 To maintain equilibrium, homeostatic regulating systems such as the one for food intake shown here provide a feedback mechanism. Low levels of glucose—the brain's primary energy source—stimulate the hypothalamus, which prompts the pancreas to release insulin and raise the levels of glucose. Once the glucose reaches a certain level, control mechanisms act to lower it.

or consistently high, the result will be an increased appetite; however, very low insulin levels lead to weight loss and very high insulin levels lead to weight gain.

CONCEPT CHECK

- Insulin levels fluctuate cyclically over the course of a day. Would you guess that they are higher in the middle of the day, when people tend to be hungry, or late at night, when they are generally less hungry? (Check your answer on page 412.)

The Long-Term Regulation of Hunger

Stomach distension, intestinal distension, and the other mechanisms for ending a meal are far from perfect. Depending on the caloric density of your next meal, you may eat a bit more or a bit less than you need to replenish your resources. If you misjudged in the same direction with every meal, you would eventually have a problem. If you consistently ate 5% more than you needed, it is estimated that you would gain 15 pounds (7 kilograms [kg]) per year (Jéquier, 1987). If you consistently ate 5% less than you needed, you would eventually starve to death.

However, you have long-term mechanisms to correct short-term errors. If you overeat for several meals, you will begin to feel less hungry and you will start eating smaller meals until you get back to about your normal weight. Conversely, if you undereat for a few meals, you will start feeling hungrier than usual, and you will increase your intake until you get back to normal. Most people's weight fluctuates from one day to the next but remains remarkably stable from one month to the next, even if they seldom check the scales. That mean weight is referred to as a **set point**—a level that the body works to maintain (Figure 11.12).

The mechanism for this correction is now partly understood. *The body's fat cells produce a hormone called leptin in amounts proportional to the total amount of fat* (Halaas et al., 1995). When the body gains fat, the extra leptin alters activity in parts of the hypothalamus (an area of the brain), causing meals to satisfy hunger faster (Campfield, Smith, Guisez, Devos, & Burn, 1995). Among the many other effects of leptin, it triggers the start of puberty: When the body reaches a certain size and weight, the increased leptin levels combine with other forces to induce all the hormonal changes of puberty (Chehab, Mounzih, Lu, & Lim, 1997).

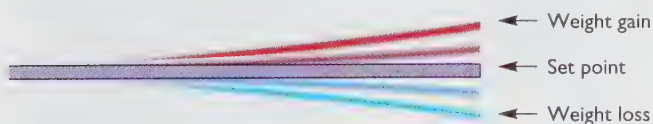


FIGURE 11.12 For most people, weight fluctuates around a set point, somewhat like a diving board that bounces up and down from a central position.

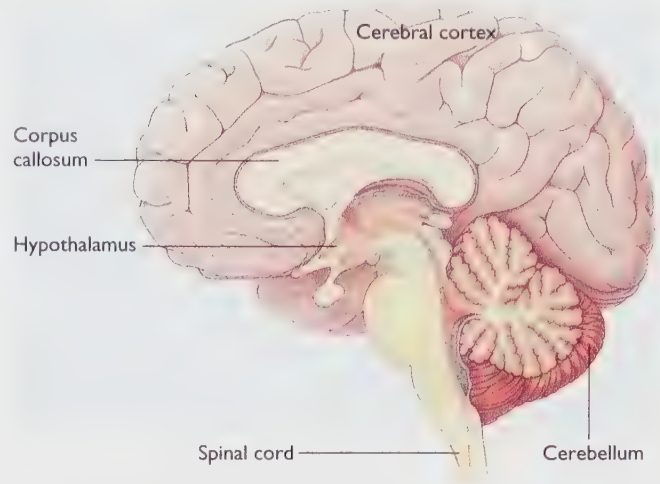


FIGURE 11.13 The hypothalamus, a small area on the underside of the brain, contains several subareas that contribute in various ways to eating, drinking, sexual behavior, and other motivated activities. Damage to the lateral hypothalamus inhibits eating; damage to other hypothalamic areas can provoke overeating.

The Brain Mechanisms of Hunger and Satiety

The brain must somehow monitor information from the blood, cells, and digestive system to determine when to eat and when to stop eating. Several brain areas, especially those in the hypothalamus (Figure 11.13), are especially important in this process.

The **lateral hypothalamus** appears to be a *critical area for starting meals*. Animals, including humans, with damage to the lateral hypothalamus have trouble salivating, swallowing, and digesting foods (Hernandez, Murzi, Schwartz, & Hoebel, 1992). Also, such individuals secrete higher levels of glucagon than of insulin; consequently, the fat supplies are constantly converted into blood glucose, so the individual has little immediate need for further nutrition.

Two areas of the hypothalamus are important for ending meals. One is the **ventromedial hypothalamus**. *After damage that includes the ventromedial hypothalamus and the axons that pass nearby, an individual digests food more rapidly than usual and secretes more insulin*. The high insulin level causes a disproportionate amount of each meal to be stored as fat. Because the food passes quickly through the digestive system and into the fat cells, where it stays, the individual becomes hungry again shortly after each meal (Hoebel & Hernandez, 1993). Figure 11.14 shows a very obese rat with this kind of brain damage. One woman with a tumor in this area gained an average of more than 10 kg (15 pounds) per month (Reeves & Plum, 1969).

Another area of the hypothalamus contributes to ending meals in a different way. *After damage to or chemical inhibition of the paraventricular hypothalamus, meals are of normal frequency but each individual meal is*



FIGURE 11.14 An obese rat with a damaged ventromedial hypothalamus (left) can eat less than an ordinary rat (right) and still gain weight. This rat's excess fat prevents it from grooming its fur.

enormous (Leibowitz & Alexander, 1991; Leibowitz, Hammer, & Chang, 1981). An individual with such brain damage can continue eating until the stomach and intestines are about to burst.

CONCEPT CHECKS

4. Many young ballet dancers and other female athletes who exercise heavily and keep their weight low are slower to start puberty than most other girls. Why?
5. After damage to the ventromedial hypothalamus, an animal's weight eventually reaches a higher than usual level and then fluctuates around that amount. What has happened to the set point? (Check your answers on page 412.)

Eating Disorders

The mechanisms I have discussed so far enable most people to select a reasonable, well-balanced diet and to maintain their weight within normal limits. In some individuals, though, the motivational mechanisms go awry. They feel hungry all the time and eat too much, or they alternate between stuffing themselves and starving themselves, or they feel hungry but refuse to eat. Some of these disorders result from physiological abnormalities; others result from social and cognitive influences that are competing with the normal physiological mechanisms.

Obesity

Obesity is the excessive accumulation of body fat. A body weight 20–40% above the standard for a person's height is considered mild obesity. Weight 41–100% above the standard is considered moderate obesity, and still greater obe-

sity is considered severe (Berkow, 1987). Why do some people become seriously overweight? Obviously, because they take in more calories than they use up. But *why* do they do that?

The Limited Role of Emotional Disturbances

Are overweight people more likely than others are to have psychological problems? One prevalent idea has been that anxiety, depression, or other emotional problems can lead to overeating and weight gain. Another theory is that extreme weight gain leads to anxiety or depression. The research, however, fails to support either hypothesis. Anxiety, depression, and other psychological concerns are not unusually common among obese people (Wadden & Stunkard, 1987). Nevertheless, there may be a subpopulation of obese people who react strongly to the unfavorable way in which they are treated. Because prejudices against the obese are so common, many obese people have a restricted social life and have trouble getting a good job. Dealing with such prejudices and barriers leads some obese people to psychological distress and low self-esteem (Friedman & Brownell, 1996).

Although emotional distress is not the cause of obesity, it can produce temporary fluctuations in eating and body



Although obesity is not significantly linked with either anxiety or depression, some obese people feel distressed and suffer from low self-esteem because of how other people react to them.

weight for almost anyone. In one survey of 100 adults (Edelman, 1981), 40 said that they overeat three or more times per month when they feel nervous, tired, lonely, or sorry for themselves. The eating binge enables them to focus their attention on eating and away from their other concerns (Heatherton & Baumeister, 1991). Such eating binges are most frequent and most extreme with people who have been dieting to lose weight (Greeno & Wing, 1994). Evidently, dieters actively inhibit their desire to eat, until a stressful experience breaks these inhibitions and releases the pent-up desire to eat.

Genetics

Obesity tends to run in families, and the weight of adopted children correlates more closely with that of their birth parents than with that of their adoptive parents (Stunkard et al., 1986). What this evidence means is that, other things being equal, certain genes increase the chances that someone will become overweight. This evidence does not mean that genetics fully explains obesity, however. In particular, consider the fact that obesity is far more common in the United States today than it was in the early 1900s or earlier. The spread of obesity is certainly not due to genetic changes, but rather to changes in lifestyle. Not that long ago, most of our ancestors were active most of the day in farming or manual labor. Today, many people drive to work, take an elevator instead of the stairs, and then sit all day in an office. Eating habits have also changed. Fast-food hamburger outlets provide inexpensive high-fat meals, restaurants with all-you-can-eat buffets encourage gluttony, and even grocery stores and supermarkets offer great varieties of very tempting high-calorie convenience foods. Most obese people show a strong preference for foods that are rich in

both fats and carbohydrates, such as cake frosting, and such foods are more readily available today than ever before (Drewnowski, 1996).

Probably the best illustration of the relationship between genetics and lifestyle for the onset of obesity comes from the Pima Indians of Arizona. At least 75 percent of Pima adults have a genetic predisposition to obesity (Thompson, Ravussin, Bennett, & Bogardus, 1997), and most also have high blood pressure and diabetes. However, before the 1940s, when they were mainly eating the vegetables that grow in the Sonoran desert, such as those shown in Figure 11.15, they were not generally obese. Evidently, their digestive systems were well adapted to that diet but not to hamburgers, French-fried potatoes, and various other convenience foods. The fact that the Pimas became obese on a “standard” U.S. diet, whereas most other people do not, points to a genetic difference. However, the impact of the change in diet indicates the importance of nongenetic factors as well. That is, a gene can either lead to obesity or not, depending on the diet. Today, many of the Pimas are trying to return to their traditional foods (Zastauri, 1996).

Decreased Energy Output

Many overweight people who claim that they eat only normal amounts of food actually eat more than they admit, maybe even more than they admit to themselves. However, some really do eat only normal-sized meals (DeLuise, Blackburn, & Flier, 1980). The overweight condition of these people is not due to high energy intake but to low energy output. Not only do they fail to exercise, but they also have a low overall metabolic rate. Presumably, one effect that the genes for obesity have is to decrease the use of energy.



FIGURE 11.15 Native Americans ate desert plants such as the beans and seeds at left. Most Pima Indians have a gene that leads to obesity if they eat a standard U.S. diet. However, few were obese when they ate their traditional diet.

One group of investigators compared the infants of 12 overweight mothers and six normal-weight mothers over their first year of life. All the babies weighed about the same at birth, but six of the babies of the overweight mothers had become overweight by the end of that first year. Those babies had also been relatively inactive since birth. During the first 3 months, they had expended about 20% less energy per day than the babies who maintained normal weight (Roberts, Savage, Coward, Chew, & Lucas, 1988).

Low energy expenditure is a good predictor of weight gain in adults as well. Eric Ravussin and his associates (1988) found that the adults with the lowest energy expenditure over a 24-hour period were the most likely to gain weight over the next 2 to 4 years.

Other Physiological Factors

People can be predisposed to obesity for many physiological reasons. For example, people who have a steadily high level of insulin store most of each meal as fat and thus become hungry soon after eating.

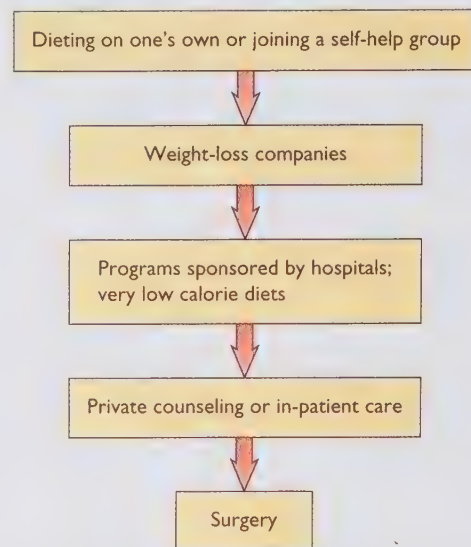
Certain genetic strains of mice become obese because their fat cells fail to produce leptin. Leptin, as you will recall, is the messenger that tells the brain how much fat the body has. If the fat cells do not produce leptin, the brain reacts as if the body had no fat—that is, as if the body were starving. Injections of leptin cause these mice to eat less, exercise more, and lose weight (Pellymouter et al., 1995). Naturally, this discovery led many people to hope that leptin injections would work similar wonders for overweight people. Indeed, a few obese people are deficient in leptin, and leptin injections would probably be a very safe and effective treatment for them (Montague et al., 1997). However, the great majority of obese people produce leptin in proportion to their fat stores, resulting in leptin levels several times larger than average (Considine et al., 1996). These people are evidently insensitive to leptin; perhaps their leptin receptors in the brain are unresponsive. Although leptin itself would be no remedy for such people, researchers may someday find another way to accomplish what leptin ordinarily accomplishes.

Losing Weight

If you talk to enough people about weight loss, some will tell you that it is almost impossible to lose weight and keep it off, but others will recount stories of people who succeeded. The reason for the discrepancy is simple: Some people do manage to lose weight and keep it off (Schachter, 1982), and then neither you nor any weight-loss clinic will hear much more about them. Those who show up at a weight-loss clinic are people who have failed repeated attempts to diet on their own, and many of them have difficult problems.

For those who seek professional help in losing weight, a variety of approaches are available. Most therapists

recommend starting with the simplest methods; if those methods fail, then consider trying more intensive methods (Friedman & Brownell, 1996):



A survey of weight-loss experts found that they disagreed about much but did agree on a few recommendations (Schwartz & Brownell, 1995):

- Except for people with unusual medical problems, almost everyone who needs to lose weight should include increased exercise as part of the strategy. The best advice is to set realistic exercise goals and then stick to them: Someone who has hardly lifted a muscle in years should not plan to run great distances every day. It is better to start with a more modest goal, such as walking three times around the block every other day, and then gradually increase the activity.
- The severity of the problem should dictate the approach. For example, only people with severe, life-threatening obesity should even consider surgical removal of fat.
- Programs such as Overeaters Anonymous (OA) are often helpful to people who feel comfortable with the strong spiritual focus and the emphasis on overeating as an addiction or disease. However, such programs should not be recommended to people who feel uncomfortable with this approach.
- Private counseling is an option mainly useful for people who have other psychological problems in addition to their weight problem.

The Effect of Intentional Weight Loss on Appetite

Are you satisfied with your own weight? In the United States, almost everyone says “no.” Even among women who have a normal weight for their height (according to life insurance tables) and who *know* that they are of normal weight, nearly all want to be thinner. Among men of normal

weight, some wish they were heavier and some wish they were thinner, but again very few are content (Raudenbush & Zellner, 1997).

The motivation for women is cultural standards that depict thin women as especially attractive or fashionable. April Fallon and Paul Rozin (1985) asked women to indicate on a diagram which body figure they thought men considered most attractive. The investigators also asked men which female figure *they* considered most attractive. As Figure 11.16 shows, women thought that men preferred thinner women than most men actually do. (The same study also found that men thought women preferred heavier men than most women actually do.)

Given the social pressure to be thin, many normal-weight people deprive themselves of food they would like to eat, keeping themselves thinner than they would have been by eating naturally (Polivy & Herman, 1987). They continue to have a strong drive to eat, however, and many factors can weaken their resolve. In one experiment, participants were told that they were taking part in market research on ice cream flavors. Some of these people were first asked to drink a milkshake; others were not. Then, they were all asked to taste three flavors of ice cream. (The dependent

variable was how much ice cream they ate.) When this experiment was conducted with people who were not dieting, participants tasted about the same amount of ice cream regardless of whether or not they had already drunk a milkshake (Figure 11.17). But when the experiment was conducted with people who were dieting, those who had first drunk a milkshake ate *more* ice cream than those who had not (Ruderman, 1986; Ruderman & Christensen, 1983). Apparently, the dieters said to themselves, “What the heck. As long as I’ve already broken my diet, I may as well eat all I want.” Not all dieters react this way when they have consumed a calorie-rich snack; as is generally the case in psychology, we can find many individual differences (Lowe, 1993). However, the point is that many dieters can be easily shaken off their diet.

Some become “yo-yo dieters,” alternating periods of weight loss with periods of weight gain. The resulting fluctuations in body weight may be harmful to health, although little research has been done to measure the degree of that risk. Consequently, authorities are not certain what advice to give to people of near-normal weight who are considering trying to lose weight. Certainly, if someone is extremely obese, then losing weight is medically important. When less obese people

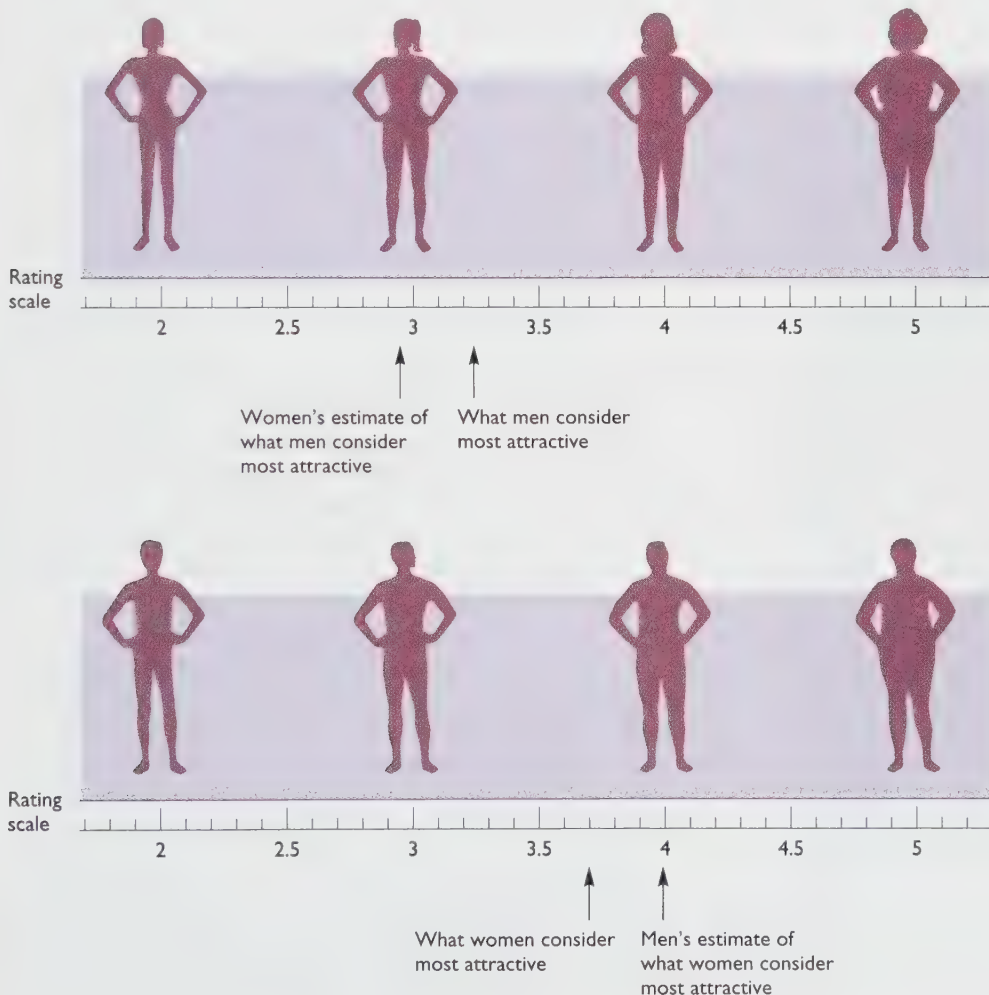
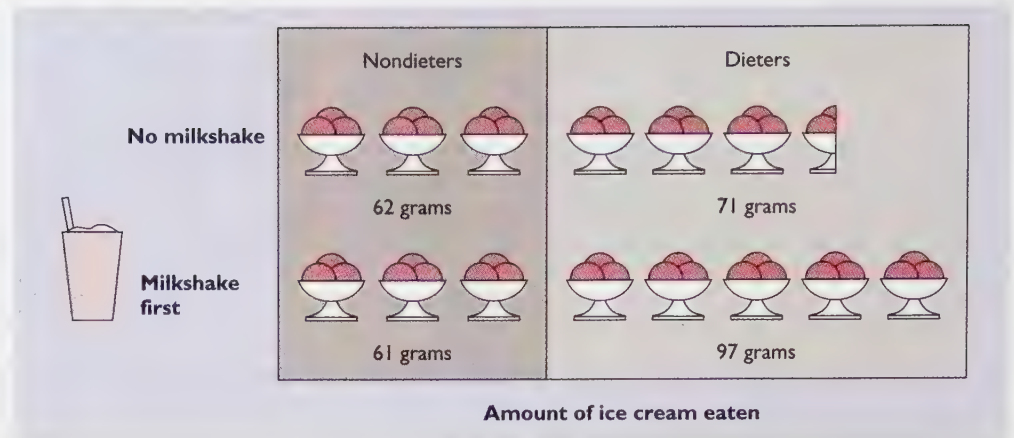


FIGURE 11.16 In a study by Fallon and Rozin (1985), women and men were asked which figure of the opposite sex they considered most attractive and which figure they believed the opposite sex considered most attractive. Each sex had systematic misestimates of the other's preferences.

FIGURE 11.17 Dieters who drank a milkshake before tasting ice cream ate more ice cream than did dieters who had not drunk a milkshake. Apparently, those who drank the milkshake thought, “I’ve broken my diet anyway, so I may as well eat all I want.” (Data from Ruderman & Christensen, 1983.)



succeed at keeping their weight down, again dieting is clearly beneficial. However, when people who are already close to normal weight find themselves repeatedly losing and regaining weight by “yo-yo dieting,” no one can be sure whether the benefits outweigh the risks (Brownell & Rodin, 1994).

Anorexia Nervosa

Some people go beyond reasonable limits in their passion to lose weight. The Duchess of Windsor once said, “You can’t be too rich or too thin.” She may have been right about being too rich, but she was definitely wrong about being too thin. Some people are so strongly motivated to be thin (for social and cognitive reasons) that they manage to overrule their physiological drives almost completely.

Here is a case history: A somewhat chubby 11-year-old girl who weighed 118 pounds (53 kg) was told to watch her weight (Bachrach, Erwin, & Mohr, 1965). She did so all through her teens. Along the way, she suffered certain hormonal difficulties, including menstrual irregularity, heavy menstrual bleeding, and deficient activity of her thyroid gland. At age 18, she still weighed 118 pounds (53 kg), but with her taller frame that weight was normal for her.

After she was married, she moved from her home in Virginia to her husband’s place of employment in California. She immediately became homesick. Because the couple could afford only a small apartment with no cooking facilities, they ate most of their meals at a very inexpensive restaurant. Soon she began to lose weight and stopped menstruating. She found sexual relations painful and unpleasant. Her physician warned her that, if she did not start regaining some weight, he would be forced to send her home to her parents. He intended this as a threat, but she took it as a promise. By the time she visited the physician again, she had lost even more weight and soon went back to Virginia.

Even after returning to familiar surroundings and home cooking, however, she continued to lose weight. The weight loss seemed to have developed a momentum of its own, and she continued to get thinner, eventually reaching a weight of only 47 pounds (21 kg).

This is a case of **anorexia nervosa**, a condition in which a person refuses to eat adequate food and steadily loses weight. (Anorexia means “loss of appetite.” Nervosa means “for nervous reasons,” to distinguish it from disorders of the digestive system.) At the outset, the person may have decided to lose weight for health reasons, or to become a dancer, or for some other reason. But the person continues losing weight well beyond the original goal. Surprisingly, though, even when anorexic women are on the verge of starvation, they have unusually high energy levels (Falk, Halmi, & Tryon, 1985). They run long distances, engage in sports, work diligently on their school assignments, and sleep very little. As with other psychological conditions, anorexia nervosa comes in all degrees, but in its most severe form it can lead to atrophy of the heart muscle and death.

Anorexia nervosa usually begins during the teenage years. Earlier onset is almost unheard of. Later onset is uncommon, although anorexia that begins during the teenage years can continue on into the 20s or beyond. Reports from before the 1980s described anorexia as almost limited to women of European ancestry. Since then, researchers have found that anorexia is indeed far more common in women than in men, although men can become anorexic too. Asian or African ancestry is no protection against anorexia; apparently, vulnerability to the disorder has far more to do with socioeconomic status and cultural pressure to be thin than with ethnic background (Le Grange, Telch, & Agras, 1997). One of the major contributors to anorexia is societal pressure, especially on women, to be very thin. Anorexia is less likely to occur in cultures that either tolerate or cherish a somewhat plumper look, such as Jamaica (Smith & Cogswell, 1994) or Europe in an earlier era (Figure 11.18).

Some psychologists have tried to explain anorexia nervosa in terms of a malfunction of the lateral hypothalamus, the brain area that seems to be so central for hunger. That explanation is unlikely, however. Many people with anorexia enjoy preparing food, seem quite interested in food, and enjoy the taste, but they avoid food in far more extreme ways than people who are merely not hungry. For example, some people with anorexia even refuse to lick